Midazolam

Mechanism of action

- Midazolam is a benzodiazepine.
- Midazolam enhances the activity of gamma-aminobutyric acid (GABA) at GABA receptors within the central nervous system, resulting in anticonvulsant activity, sedation, amnesia, anxiolysis and muscle relaxation.

Authorised practice levels

- Paramedics:
  - IM for acute behavioural disturbance, seizures and care during the last days of life.
  - IV for seizures and care during the last days of life.
- ICPs and CCPs: all routes and all indications.

Indications

- ✓ Generalised seizures that continue for more than five minutes, or seizures are recurrent.
- ✓ Acute behavioural disturbance causing a mild to moderate risk to safety and droperidol is unavailable or ineffective.
- ✓ Pain associated with severe muscle spasm or severe anxiety if adequate analgesia is not being achieved with an opiate and ketamine is not appropriate.
- ✓ Sedation, for example, for joint relocation.
- ✓ Severe anxiety associated with COPD.
- ✓ Control of anxiety or shortness of breath that does not respond to an opiate during the last days of life.

Contraindications

- ✗ Known severe allergy.
- ✗ Aged less than 12 years (analgesia only).

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Cautions

- ⚠ Altered level of consciousness. Midazolam will further reduce the level of consciousness.
- ⚠ Respiratory depression or at high risk of respiratory depression. For example, severe COPD, morbid obesity or on home BiPAP. Such patients may develop respiratory depression following midazolam administration.
- ⚠ Signs of shock. Midazolam may worsen shock.
- ⚠ Concurrent administration of opiates or ketamine. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of midazolam will be increased and prolonged in this cohort.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but midazolam should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- The dose of midazolam is dependent on the indication and the route. See the individual guidelines.
- If a dose range is described, use a dose at the lower end of the range if a caution is present.
- If midazolam is being administered as an adjunct to analgesia or sedation, the patient must be rousable at all times.

Administration

- IV administration:
  - Dilute 2 ml from a 15 mg/3 ml ampoule to a total of 10 ml using 0.9% sodium chloride. This solution contains 1 mg/ml.
  - Administer IV as a bolus.
- IM administration:
  - Draw up the dose from a 15 mg/3 ml ampoule. Do not dilute.

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- The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Sedation.
- Respiratory depression.
- Hypotension.
- Amnesia.

Usual onset of effect

- IV: 2-3 minutes.
- IM: 3-5 minutes.

Usual duration of effect

- 30-60 minutes. The sedative effect may be longer, particularly in the elderly.

Usual preparation

- Ampoule containing 15 mg in 3 ml.

Pharmacokinetics

- Midazolam is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects will be increased and prolonged in the presence of other sedatives or pain relieving medicines (for example, other benzodiazepines, opiates, ketamine and alcohol).

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More Information

Version history

9 June 2025 1.0.5.3

Update: Indications updated.

11 September 2023 1.0.4

Update: Indications have been updated to align with changes to EAS CPG 5.1 and 5.8.

22 November 2022 1.0.1

Update: Contraindications, cautions and aligned with analgesia and end of life CPGs.